# HARSHITH NARASIMHAMURTHY

**Data Scientist | ML Engineer | Generative AI**

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## PROFESSIONAL SUMMARY

Data Scientist with 3.5+ years building production ML systems across fintech, insurance, and agribusiness. Delivered 5 end-to-end ML products from geospatial credit scoring (~90% accuracy) to multi-agent GenAI systems with RAG and fine-tuned LLMs. Proven across the full lifecycle: feature engineering, model deployment (FastAPI/Docker), MLOps (Airflow/MLflow), and CI/CD on AWS.

## PROFESSIONAL EXPERIENCE

**Data Analyst 2** | SatSure Analytics, Bengaluru, India *Jul 2023 – Present*

- **Farmer Credit Score System :** Architected and deployed an end-to-end Farmer Credit Scoring system leveraging geospatial data, satellite imagery, and Random Forest models (~90% accuracy) to quantify creditworthiness and determine optimal lending amounts. Currently adopted by 5+ national banks including HDFC, ICICI, IDFC First, and L&T Finance, generating ~₹12L+ monthly recurring revenue and accelerating agricultural loan underwriting cycles by ~25%.
- **Village Ranking | Location Intelligence Product :** Designed and developed Village Ranking, a village-level location intelligence product integrating real-time agricultural, demographic, and infrastructure data to drive strategic decisions, including bank branch expansion and optimized distribution of seeds, fertilizers, and agricultural machinery for enterprise clients such as Syngenta and Bayer.
- **Geospatial Data Products & Business Impact :** Developed and scaled a portfolio of geospatial intelligence products, including Crop Health Monitoring, Regional Agricultural Performance Ranking, Cropping Intensity, Irrigation Assessment, and Revenue Estimation, enabling adoption across business verticals and client organizations and contributing to new revenue streams and cross-selling opportunities.
- **Geospatial Validation & Model Accuracy :** Built a robust geostatistical validation framework combining mathematical models with government ground-truth datasets to evaluate product accuracy across geographies and growing seasons throughout India. Achieved 92%+ overall accuracy across primary data products, validated against real-world agricultural events, including droughts, floods, and biotic stresses. Established the framework as the company-wide standard for product validation and reliability.
- **Machine Learning, GenAI & Intelligent Automation :** Applied classical ML techniques, including Random Forests, Decision Trees, and regression, alongside GenAI approaches involving RAG pipelines, transformer-based solutions, and agentic workflows using LangChain and LangGraph. Contributed to internal AI-powered tooling that reduced turnaround time for ad hoc data requests by approximately 40%.
- **End-to-End Data Engineering & Cloud Deployment :** Owned the end-to-end data lifecycle, spanning data acquisition, experimentation, feature engineering, modeling, validation, and production deployment using Docker, Kubernetes, and AWS services (S3, Athena, and EC2). Translated complex geospatial and agricultural data into actionable stakeholder insights through Tableau dashboards.
- **Technical Leadership & Project Management :** Mentored 3+ interns and currently guide 2 junior data analysts in end-to-end project execution, technical problem-solving, and delivery. Managed project lifecycles and cross-functional collaboration using Jira, driving timely delivery and stakeholder alignment. Recognized as a Customer Champion at SatSure for excellence in project execution.

**ML Research Intern** | Western Sydney University, Sydney, Australia *Jan 2023 – Mar 2023*

- Developed LSTM & ARIMA time-series forecasting models for crop yield prediction; engineered preprocessing and cross-validation workflows improving forecasting accuracy by ~12% over baseline.

## KEY ML & GENAI PROJECTS

**NovaPay — Multi-Agent AI Customer Support System** | *Gemma-3 4B, QLoRA, RAG, LangChain, HuggingFace Spaces*

- Architected multi-agent orchestration with QLoRA fine-tuned triage routing to RAG-grounded specialists; scored 4.9/5.0 on LLM-as-Judge and handled 15/15 adversarial attacks, deployed live on HF Spaces at $0 infra cost.
- Fine-tuned Gemma-3 4B (0.76% trainable params, 15 min on free T4) on 884 synthetic examples with quality-controlled data generation; built hallucination detection and provider-agnostic LLM client supporting 4 backends.
- Designed end-to-end evaluation: A/B/C comparison, BERTScore (F1 = 0.74), adversarial red-teaming, latency profiling, and 7-KPI ops dashboard; rigorously validated RAG's 10.6% faithfulness improvement over baseline.

**Formula 1 AI SQL Assistant — LLM-Powered Text-to-SQL Engine** | *Groq LLaMA 3.3 70B, LangChain, FastAPI*

- Built production Text-to-SQL agent with conversation memory, self-correction logic, and SELECT-only guardrails; enables zero-SQL natural language querying over live structured F1 race data via REST API.

**Customer Segmentation MLOps Pipeline — End-to-End ML System** | *KMeans, MLflow, DagsHub, FastAPI, Streamlit, Docker*

- Architected full MLOps lifecycle for RFM + behavioral customer segmentation; tracked experiments via MLflow + DagsHub for reproducible model comparison across runs.
- Deployed via FastAPI + Streamlit, containerized with Docker, automated CI/CD to Docker Hub via GitHub Actions; modular ingestion → validation → transformation → training pipeline from MongoDB with pytest coverage. Live on Streamlit Cloud.

**Real-Time Crop Health Monitoring — Production Satellite ML System** | *NDVI, K-Means, Geospatial ML, AWS*

- Deployed 10m × 10m NDVI time-series platform with K-Means health classification (Good/Average/Poor); achieved ~95% damage-detection accuracy and adopted by banks and insurers for loan underwriting and yield assurance at scale.

## TECHNICAL SKILLS

**Languages:** Python, SQL, Bash
**ML / DL:** Scikit-learn, PyTorch, Transformers, PEFT/QLoRA, LSTM, Random Forest, XGBoost, Clustering, A/B Testing
**Generative AI:** LLMs, RAG, LangChain, HuggingFace, Prompt Engineering, Agentic AI, Fine-tuning
**MLOps & Cloud:** MLflow, Apache Airflow, Docker, GitHub Actions CI/CD, AWS (S3, Athena, EC2, ECR), FastAPI, Streamlit
**Data & BI:** Pandas, NumPy, Apache Iceberg, MongoDB, ETL/ELT, Tableau, Matplotlib, Seaborn

## EDUCATION & RECOGNITION

- M.Sc. Agricultural Statistics | Anand Agricultural University | 2021–2023 | Gold Medalist (University Rank #1).
- B.Sc. Agriculture | University of Agricultural Sciences, Bengaluru | 2017–2021 | All India Rank #6.
- Certifications: Generative AI with LangChain & HuggingFace • HackerRank SQL (Advanced).
- Publications: Memory-based neural network for cumin price forecasting • RNN architecture for banana price forecasting.